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Nitte (Deemed to be University)

(Established under Sec.(3) of UGC ACT, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25CCT101 – CONSTRUCTION PLANNING AND CONTROL MANAGEMENT

(For CCT)

Duration: 3 Hours]

[Max. Marks: 100

Note: 1. Answer **Five full** questions choosing **One full** question from each **Unit**.

2. Usage of Normal distribution table is permitted.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Emphasize on the importance of an organization for any establishment. | 8 | L2 | 1 | 1 |
| b) Explain "Functional Organization Structure" with the help of flow chart and list the merits and demerits. | 12 | L2 | 1 | 1 |
| 2. a) Highlight the advantages and disadvantages of bar chart. | 8 | L1 | 1 | 1 |
| b) Draw the network representing the following activity relationship using minimum number of dummies and number the nodes as per Fulkerson's rule.
A, B and C are the initial activities
Q, M and R the finishing activities.
H precedes N but follows D
E follows D but precedes N
F and L follow K and B
F precedes M
Q follows P, L and N
L precedes R but follows C
D precedes G
K and D follow A
F precedes P but follows G | 12 | L3 | 1 | 3 |

Unit – II

3. a) State the situations of applications of PERT and CPM in project management and enumerate the differences between them.
- b) The following table gives the activities in a construction project and their durations.

Activity	1-2	1-3	1-4	2-5	3-5	3-6	4-7	6-7	5-8	6-8	7-8
Duration (Weeks)	7	8	4	3	9	10	5	4	2	8	4

Draw the network, Determine the critical path, Find out Earliest times, Latest times, Total float and Free float for each activity.

4. a) Draw the arrow network diagram for the following details and number the events in tens as per Fulkerson's rule.

Activity	A	B	C	D	E	F	G	H	I	J
Preceded by	-	A	B	C	A	C,E	D,F	E,C	F,H	G,I

- b) Explain the following with relevant sketches and equations.
i) Total float ii) Free float ii) Independent float

08	L3	2	2
12	L2	2	1



Unit – III

5. a) Explain the following relating to PERT
i) Expected time of completion (tE) ii) Pessimistic time (tP)
iii) Probability factor (Z) iv) Normal distribution curve

08 L2 3 2

- b) Dates regarding the activities in a project are given below.

Activity	1-2	1-3	1-4	2-6	3-7	4-5	5-7	6-7	5-8	6-8	7-8
Optimistic	18	12	13	10	11	14	17	16	19	15	18
Most likely	22	14	17	13	15	18	21	18	24	18	19
Pessimistic	26	19	22	17	18	23	23	21	27	24	21

- i) Draw the project network, Determine the expected duration of all activities
iii) What is expected project length?
iv) What is the probability that the project will be completed at least 3 weeks earlier than and 3 weeks later than the expected project duration?

12 L3 3 3

6. a) Write short notes on following:

i) Risk Management ii) Risk Register

08 L2 3 1

- b) Write short note on following by highlighting their relevance in project management:

i) Resource Allocation ii) Resource Leveling iii) Updating Schedule

12 L2 3 1

Unit – IV

7. a) Discuss time-cost optimization in project management

08 L4 4 1

- b) The following table gives data for a simple construction project. Crash critical activities of the project to maximum possible extent, show the time scaled version and hence determine the cost and duration at each stage of crash.

Activity	Time (Days)		Cost	
	Normal	Crash	Normal	Crash
10-20	10	8	100000	1200000
10-30	7	5	90000	105000
20-30	4	3	70000	83000
20-40	9	7	60000	72000
30-40	7	6	50000	61000

Consider indirect cost per day as Rs 2000.

12 L5 4 4

8. a) Solve the following LPP using Simplex Method for optimal solution.

Minimize $Z = X_1 - 3X_2 + 2X_3$

Subject to constraints

$$3X_1 - X_2 + 2X_3 \leq 7$$

$$-2X_1 + 4X_2 \leq 12$$

$$-4X_1 + 3X_2 + 8X_3 \leq 10$$

08 L4 4 3

- b) Solve the following L P Problem by graphical method.

Maximize $Z = 13X_1 + 23X_2$

Subject to $5X_1 + 15X_2 \leq 480$

$$4X_1 + 4X_2 \leq 160$$

$$35X_1 + 20X_2 \leq 1190$$

$$X_1 \geq 0, X_2 \geq 0$$

12 L4 4 3

Unit – V

9. a) How is economic order quantity evaluated? Explain with the help of suitable equations.

08 L2 5 1

- b) Write short notes on the following indicating salient features.

i) ABC analysis

ii) Factors to be considered in the process of Material Management.

12 L2 5 1

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|--------|--|----|----|---|---|
| 10. a) | Explain the objectives and advantages of Inventory control. | 08 | L2 | 5 | 2 |
| b) | A company makes bicycles. It produces 450 bicycles in a month. It buys the tires for bicycles from a supplier at a cost of Rs 20 per tire. The company's inventory carrying cost is estimated to be 15% of cost and the ordering is Rs 50 per order. Calculate the following | | | | |
| | i). E O Q | | | | |
| | ii) Number of orders per year | | | | |
| | iii) Average annual ordering cost | | | | |
| | iv) Average inventory | | | | |
| | v) Average annual carrying cost | | | | |
| | vi) Total cost | 12 | L3 | 5 | 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
